A headline goes here.

Another lightly psychotic document that's supposed to resemble an academic report by

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- A neurotic guy

A retrospect of a 5-week introductory course to data structures.
Looking back on the assignments from these past weeks, most of them seem so simple now. But before I got to this point, I had a lot of headaches, self doubts, rethinking if this programme really is something for me. I'm still not entirely sure of anything, but I do know that I feel a lot more confident around the data structures we have gone through (although, at the point of writing this I am not finished with my tree structure…) and I feel like I could, at least, write pseudocode for all of these structures and algorithms. There's lot I'm still unsure about regarding programming, like how my classes and scripts communicate with each other, the right way to structure and abstract my scripts, but every day I feel that I get a little bit closer to becoming capable of one day being mistaken for a programmer by someone who have never touched a computer….

… 5 weeks earlier...

Week 1, Stacks upon stacks, upon stacks.

The week that set it all off. I remember being very confused this week, having coming off the vampire survivors course, into this one, and my first task is to implement a stack. And I get an entire week to do it. Having been so stressed with my vampire survivors for the last few weeks, this change of pace was making me even more stressed for some reason.

As has been the theme for all of these weeks, I spent the first few days confused regarding the assignment, the data structure. A lot of people in my class had a hard time grasping the concepts of stacks and queues, I had an edge here since I studied programming I during the summer, where I just followed the curriculum of a course book in C# for a week before I could write a passing grade on the test. So I was very familiar with the concepts of these basic structures, but for some reason I just couldn't wrap my head around defining my own stack. That just seemed so foreign to me for some reason. But by the end of the week I had defined and implemented my Stack2TheStreets, for the code review I got to define a queue, which I named Wiiwoo, and at that point it just really felt super easy and I ended that week on a high, feeling like the rest of this course would be just as easy and let my newfound confidence get the best of me.

But what he didn't know was that JSON was just around the corner waiting for him...

Week 2. JAAAAAAASOOON

This week was tough. Rising so high during the previous week Just to have Json grab hold of me and send me spiraling down to the ground.

We worked with Json during our Vampire survivors project, so I knew how jsonutilities worked, and the basic workflow of writing to, and reading from, Json. So this should be easy, right? We are just reading, not writing to anything. Should be like, half the work, right?

No.

I can't even explain why this week whooped me so hard. I guess it was the combination of json, scriptable objects, and having a really hard time with our "tasks" not *feeling like real* tasks.

I probably spent the better part of the week making a UI for the computer to make it feel more like I'm receiving, and, completing tasks, and adding debugtasks of my own just to have it feel more like there's something being done.

Oh right, a large part of the problems I would have with the later weeks was created during this week. With my Employees-class, and a class called EmployeeList, which just automagically made a list of Employees for me, I thought. More on that later.

Week 3 – WHY WON’T IT WORK

This week I think I got a head start on making the code. By wednesday I had written my merge sort function, and I had it working on ints. But I couldn’t for the life of me get it to work with my EmployeeLists. I spent so much time just staring into the computer screen, being so sick of coding, codes, numbers, letters.

But the problem was of course not with my impeccable coding with the merge sort. The problem was with my EmployeeList-class, and me not initializing the Employees in said list correctly. We discovered this when we sat 3 people around my computer after I had written the quick sort, and it still wouldn’t work.

That made me feel really stupid.

Anyways.

The quick sort was really quick. It frose my computer for 5 minutes and 5 seconds.

The problem somehow lied in the initializer inside of the Employees-class… Many hours of troubleshooting and testing random stuff and it finally worked. It was fast for real! Having written both codes, I think I prefer the merge sort. It has better vibes.

Week 4 – maybe I should switch to game design…?

Not because this week was any harder than the previous ones. Just that I spent much more time this week on making an interface on the computer in the game to show the span of employees, an interface to enter two dates. And for some reason defining how my search should handle DateTimes, when there are built in tools for that stuff.

But we could just say that I did it for the love of the game.

I actually spent some time on optimizing my search also. Having my binary search being recursive and writing lists with every recursion, to being in a simple while loop and just returning the indices of the given employees in the list, and later printing them onto the computer screen in the game.

It went from 70K ticks down to 40, not K, 40 ticks. That really made me feel like I some day might shape up to be something of a programmer. At the moment of writing this, the search button has stopped working in the UI in the game. I really need to fix that. But I’m in procrastination mode right now and I’d much rather continue to make this document many, many times longer than it needs to be.

Week 5, anxiety starting to kick in

GP1 about to start, just heard a couple of hours ago that we won't have any time to work on this project on friday, so i'll just have to do my best to finish this by the end of tomorrow, thursday. Unless I feel that I could squeeze another 1100 words into this document.

I feel that trees should be super simple. It's just such a simple concept (talking BST here)

You put stuff into it. Stuff smaller than the root goes to the left, stuff bigger goes to the right.

So why is it so impossible for me to figure out how this works.

I have read several pseudocode, and actual code-implementations of this structure.

But what's wrinkling my brain so much I guess is that you got the node, right? But that node contains nodes. And those nodes sure as heck has nodes aswell. And i'm supposed to send this mess of nodes between scripts to have them searched through in a separate task.

It's like, I know that this shipment of code won't get scuffed during the delivery, but I still just have a hard time understanding that my task will know what to do with this when I send it. That it will start at the right node, and that another mystery node has somehow become the root. It's just so hard. I'f you're Jo reading this, I probably have messaged you being really confused already about this. If I still have the energy after writing this I will write to you on teams about this.

And like all earlier weeks, I feel like I would have quite an easy time of explaining how BSTs and DFS works, and write pseudocode. But actually implementning it and wrapping my head around the implementation, abstraction, abracadabras and all of that.

Anyway, I have written all of the code for the tree, inserting into the tree, searching the tree, and printing the search to the console.

Oh, and Alfred Pugsley did it. I saw it with my own eyes. That bugger. If I'm wrong, I refuse to believe it.

Bonus Chapter - ..

- …

I can’t think of a funny name for this one.

If you’ve stuck with me to the end I want to thank you for giving me the time of day for reading this document. My report from the vampire survivors course was a hot mess, I forgot to add titles, it was much more like a stream of consiousness than an academic document. I’m not an academian, and that should be very clear by the contents and structure of this document. I believe that the headaches and strain of learning and implementing these data structures will one day lead to me having a broad understandning of this field, and that it will help me, if I choose to continue down this path of becoming a part of the game industry, to make some really good games.

So while I have remembered to add titles to this document, I have still followed the same process in that I sat down with a blank document, and won’t stand up until I have a completed document, and that said document will ooze with my lack of professionalism and personality.

Aight. Thanks for reading, have a great day. Game tip of the week, Queens Domain. Just trust me, you won’t regret it. I don’t know your taste in games, but I can’t imagine anyone disliking such a beautifully crafted game. And it’s massive! 26 hours in and I would guess that I still have many hours to go before I’m done. You would never guess it was made by 2 people.